

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q24-2421

Function Junction 100 kV  
September 23, 2025

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2421. The Interconnection Customer proposes a 95.3 MW battery energy storage facility interconnecting at the Function Junction 100 kV substation in the Ohio area. The request was received on June 18, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 415 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 32 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$107,912,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Unknown12262 Substation – W Lancaster 100 KV  
Contingency: Omni-Hackberry – Norwich 500 KV  
Loading: 157.8% (Rating: 149 MVA)  
DFAX: 7.9%  
Pre-project loading: 152.8%
2. Facility: Wst Lafayette – Unknown12262 100 KV  
Contingency: Omni-Hackberry – Norwich 500 KV  
Loading: 148.6% (Rating: 161 MVA)  
DFAX: 7.9%  
Pre-project loading: 143.9%
3. Facility: West Lancaster – Choupique 100 KV L  
Contingency: Omni-Hackberry – Norwich 500 KV  
Loading: 147.8% (Rating: 133 MVA)  
DFAX: 7.9%  
Pre-project loading: 142.1%
4. Facility: Gray Summit – Function Jct 100 KV  
Contingency: Tecumseh Rd Substation – Rio Bravo 100 KV L  
Loading: 136.3% (Rating: 116 MVA)  
DFAX: 70.0%  
Pre-project loading: 78.6%
5. Facility: West Lafayette – Mossyrock 100 KV  
Contingency: Omni-Hackberry – Norwich 500 KV  
Loading: 135.5% (Rating: 183 MVA)  
DFAX: 7.9%  
Pre-project loading: 131.4%

6. Facility: Choupiue – Goldtown 100 KV  
Contingency: Norwich – Omni-Hackberry 500 KV  
Loading: 132.2% (Rating: 119 MVA)  
DFAX: 7.9%  
Pre-project loading: 125.8%
7. Facility: Du Pont Substation – Madison West 100 KV line  
Contingency: Brisbois – W Memphis 100 KV  
Loading: 130.6% (Rating: 128 MVA)  
DFAX: 8.8%  
Pre-project loading: 124.0%
8. Facility: Madison W – Speen St 100 KV  
Contingency: Brisbois – West Memphis 100 KV L  
Loading: 126.6% (Rating: 159 MVA)  
DFAX: 8.8%  
Pre-project loading: 121.3%
9. Facility: Goldtown – Sibley No Two Substation 100 KV  
Contingency: Snipes – Trimble 100 KV L  
Loading: 123.1% (Rating: 112 MVA)  
DFAX: 43.4%  
Pre-project loading: 86.1%
10. Facility: Keelers – Poudre 100 KV  
Contingency: Function Junction – Gray Summit 100 KV  
Loading: 121.8% (Rating: 92 MVA)  
DFAX: 100.0%  
Pre-project loading: 17.9%
11. Facility: Trimble – Snipes 100 KV  
Contingency: Sibley No Two – Goldtown 100 KV  
Loading: 120.4% (Rating: 152 MVA)  
DFAX: 58.0%  
Pre-project loading: 84.2%
12. Facility: Daly City Tap – Speen St 100 KV  
Contingency: Brisbois – West Memphis 100 KV  
Loading: 118.9% (Rating: 195 MVA)  
DFAX: 8.8%  
Pre-project loading: 114.6%
13. Facility: Lake Rd – N Gila 100 KV  
Contingency: Daly City Tap – Speen Street 100 KV  
Loading: 118.7% (Rating: 132 MVA)  
DFAX: 5.8%  
Pre-project loading: 114.5%
14. Facility: Lakerd – Mwd Etiwanda 100 KV  
Contingency: Speen St – Daly City Tap 100 KV  
Loading: 117.7% (Rating: 141 MVA)  
DFAX: 5.8%  
Pre-project loading: 113.8%
15. Facility: Gray Summit – Trimble 100 KV  
Contingency: Sibley No Two – Goldtown Substation 100 KV line  
Loading: 117.1% (Rating: 151 MVA)  
DFAX: 58.0%  
Pre-project loading: 80.4%
16. Facility: W Memphis – Mwd Etiwanda 100 KV  
Contingency: Speen St – Daly City Tap Substation 100 KV

Loading: 112.9% (Rating: 163 MVA)

DFAX: 5.8%

Pre-project loading: 109.5%

17. Facility: Goose Lake – Jug St 100 KV Line

Contingency: W Memphis – Brisbois 100 KV Line

Loading: 108.0% (Rating: 191 MVA)

DFAX: 5.2%

Pre-project loading: 105.4%

18. Facility: Keelers Substation – Chaffins 100 KV

Contingency: Gray Summit Substation – Function Jct 100 KV

Loading: 107.9% (Rating: 115 MVA)

DFAX: 100.0%

Pre-project loading: 24.9%

19. Facility: Miami Dade County Resource Recovery Fac – Labarre 100 KV

Contingency: Myrtle-Date – Waterford Power Llc 230 KV Line

Loading: 107.5% (Rating: 107 MVA)

DFAX: 7.7%

Pre-project loading: 100.7%

20. Facility: Function Jct – Poudre 100 KV

Contingency: Rio Bravo – Tecumseh Rd 100 KV

Loading: 105.5% (Rating: 109 MVA)

DFAX: 30.0%

Pre-project loading: 79.3%

21. Facility: Brisbois – W Memphis 100 KV

Contingency: Daly City Tap – Speen St 100 KV L

Loading: 105.5% (Rating: 213 MVA)

DFAX: 5.8%

Pre-project loading: 102.9%

22. Facility: Mead Pump Substation – Magdalena 100 KV Line

Contingency: West Memphis – Brisbois 100 KV

Loading: 104.9% (Rating: 154 MVA)

DFAX: 5.2%

Pre-project loading: 101.6%

23. Facility: Port Washington Sw Yd – Chaffins 100 KV

Contingency: Christenson – Mechanicville Hydroelectric Sta 100 KV

Loading: 103.2% (Rating: 121 MVA)

DFAX: 20.8%

Pre-project loading: 86.7%

24. Facility: Smithers – Mechanicville Hydroelectric Station 100 KV Line 1

Contingency: Mechanicville Hydroelectric Sta – Smithers 100 KV line

Loading: 102.0% (Rating: 271 MVA)

DFAX: 12.6%

Pre-project loading: 97.6%

25. Facility: Smithers – Mechanicville Hydroelectric Sta 100 KV Ckt 2

Contingency: Mechanicville Hydroelectric Sta – Smithers 100 KV

Loading: 102.0% (Rating: 271 MVA)

DFAX: 12.6%

Pre-project loading: 97.6%

26. Facility: Filer – Tidd 500 KV

Contingency: Lerdo – Tidd 500 KV

Loading: 101.7% (Rating: 2,413 MVA)

DFAX: 7.3%

Pre-project loading: 101.4%

27. Facility: Russell Fork – Robison Park 230 KV

Contingency: Dekalb – Jones Chapel 500 KV

Loading: 101.5% (Rating: 4,142 MVA)

DFAX: 9.3%

Pre-project loading: 101.3%

28. Facility: Robson Park – Dekalb 230 KV L

Contingency: Jones Chapel – Deklb 500 KV Line

Loading: 101.5% (Rating: 4,498 MVA)

DFAX: 6.8%

Pre-project loading: 101.3%

29. Facility: Apopka South – Statesville Delivery No 3 100 KV

Contingency: Christenson – Mechanicville Hydroelectric Station 100 KV Line

Loading: 101.4% (Rating: 143 MVA)

DFAX: 5.7%

Pre-project loading: 97.6%

30. Facility: Port Washington Sw Yd – Labarre 100 KV L

Contingency: Statesville Delivery No 3 – Apopka South 100 KV

Loading: 101.2% (Rating: 138 MVA)

DFAX: 14.7%

Pre-project loading: 91.1%

31. Facility: Medora – P.T.S 345 KV

Contingency: Ritter Park – Sugar Hill 345 KV L

Loading: 101.2% (Rating: 2,278 MVA)

DFAX: 8.2%

Pre-project loading: 100.9%

32. Facility: Merret – Sunrise Center 500 KV

Contingency: Wahiawa – N Allegheny Windpower Project 345 KV

Loading: 101.2% (Rating: 2,736 MVA)

DFAX: 5.7%

Pre-project loading: 101.0%

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Unknown12262 Substation – W Lancaster 100 KV | Rebuild line with higher-capacity conductor | \$5,846,000   | 100% (\$5,846,000)    |
| Wst Lafayette – Unknown12262 100 KV          | Rebuild line with higher-capacity conductor | \$7,328,000   | 100% (\$7,328,000)    |
| West Lancaster – Choupique 100 KV L          | Rebuild line with higher-capacity conductor | \$3,747,000   | 100% (\$3,747,000)    |
| Gray Summit – Function Jct 100 KV            | Rebuild line with higher-capacity conductor | \$3,915,000   | 78% (\$3,040,000)     |

| Facility                                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Mossyrock – W Lafayette 100 KV                               | Rebuild line with higher-capacity conductor | \$14,552,000  | 100% (\$14,552,000)   |
| Choupiue – Goldtown 100 KV                                   | Rebuild line with higher-capacity conductor | \$4,702,000   | 100% (\$4,702,000)    |
| Madison West – Du Pont 100 KV line                           | Rebuild line with higher-capacity conductor | \$1,286,000   | 100% (\$1,286,000)    |
| Madison W – Speen St 100 KV                                  | Rebuild line with higher-capacity conductor | \$2,650,000   | 100% (\$2,650,000)    |
| Goldtown – Sibley No Two Substation 100 KV                   | Rebuild line with higher-capacity conductor | \$7,792,000   | 100% (\$7,792,000)    |
| Keelers – Poudre 100 KV                                      | Rebuild line with higher-capacity conductor | \$5,070,000   | 100% (\$5,070,000)    |
| Trimble – Snipes 100 KV                                      | Rebuild line with higher-capacity conductor | \$2,832,000   | 73% (\$2,056,000)     |
| Speen St – Daly City Tap 100 KV                              | Rebuild line with higher-capacity conductor | \$3,313,000   | 100% (\$3,313,000)    |
| Lake Rd – North Gila 100 KV                                  | Rebuild line with higher-capacity conductor | \$5,654,000   | 100% (\$5,654,000)    |
| Lakerd – Mwd Etiwanda 100 KV                                 | Rebuild line with higher-capacity conductor | \$13,769,000  | 100% (\$13,769,000)   |
| Gray Summit – Trimble 100 KV                                 | Rebuild line with higher-capacity conductor | \$4,662,000   | 73% (\$3,385,000)     |
| W Memphis – Mwd Etiwanda 100 KV                              | Rebuild line with higher-capacity conductor | \$2,701,000   | 100% (\$2,701,000)    |
| Goose Lake – Jug St 100 KV                                   | Reconductor line                            | \$1,248,000   | 100% (\$1,248,000)    |
| Chaffins – Keelers 100 KV Line                               | Reconductor line                            | \$3,099,000   | 100% (\$3,099,000)    |
| Labarre – Miami Dade County Resource Recovery Fac 100 KV L   | Reconductor line                            | \$3,357,000   | 100% (\$3,357,000)    |
| Function Jct – Poudre 100 KV                                 | Reconductor line                            | \$3,251,000   | 60% (\$1,944,000)     |
| Brisbois – W Memphis 100 KV                                  | Reconductor line                            | \$1,707,000   | 100% (\$1,707,000)    |
| Mead Pump Substation – Magdalena 100 KV Line                 | Reconductor line                            | \$657,000     | 100% (\$657,000)      |
| Port Washington Sw Yd – Chaffins 100 KV                      | Reconductor line                            | \$4,121,000   | 46% (\$1,882,000)     |
| Mechanicville Hydroelectric Sta – Smithers Substation 100 KV | Reconductor line                            | \$1,066,000   | 19% (\$198,000)       |
| Smithers – Mechanicville Hydroelectric Sta 100 KV Ckt 2      | Reconductor line                            | \$1,256,000   | 19% (\$234,000)       |
| Filer – Tidd 500 KV line                                     | Reconductor line                            | \$21,390,000  | 1% (\$206,000)        |
| Robison Park – Russell Fork 230 KV                           | Reconductor line                            | \$18,225,000  | 1% (\$183,000)        |
| Robson Park – Dekalb 230 KV L                                | Reconductor line                            | \$10,261,000  | 1% (\$101,000)        |
| Statesville Delivery No 3 – Apopka S 100 KV                  | Reconductor line                            | \$4,296,000   | 100% (\$4,296,000)    |
| Labarre – Port Washington Sw Yd 100 KV L                     | Reconductor line                            | \$3,294,000   | 45% (\$1,471,000)     |

| Facility                       | Required Upgrade           | Cost Estimate | Allocation to Project |
|--------------------------------|----------------------------|---------------|-----------------------|
| P.T.S – Medora 345 KV          | Reconductor line           | \$11,672,000  | 1% (\$118,000)        |
| Merret – Sunrise Center 500 KV | Reconductor line           | \$32,493,000  | 1% (\$320,000)        |
|                                | Total allocated to project |               | \$107,912,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.